The output voltage of a half-wave rectifier is calculated using the relationship $\frac{E_{max}}{\pi}$

Where E_{max} is the maximum voltage of the input and π is the phase angle (180°)

$$Ripple\ factor = f$$

Where f is the frequency.

$$Efficiency = 40.6\%$$

4.5 FULL-WAVE RECTIFIER

Full-wave rectifier rectifies the full cycle of the input AC waveform, i.e., it rectifies both the positive and negative cycles of the input waveform.

If a transformer with a centre-tapped secondary winding is used, the centre-tapped secondary produces two anti-phase outputs as shown in Figure 4.11.

In the centre-tapped full wave rectifier, two diodes are used. These are connected to the centre-tapped secondary winding of the transformer. Centre-tapped winding divides the total secondary voltage into equal parts.

The circuit diagram of full-wave rectifier is shown in Figure 4.11(a). The primary winding of the centre-tapped transformer is applied with the A.C Voltage. The two diodes D_1 and D_2 are connected to the secondary winding of the transformer. These diodes D_1 and D_2 conduct both the positive and negative half cycle of the input waveform.

For the positive half cycle of the input, diode D_1 is connected with upper end (A) positive terminal and diode D_2 is connected with lower end (B) negative terminal. Thus, the diode D_1 is in forward bias and the diode D_2 is in reverse bias. As the diode D_2 is in reverse bias, it does not conduct. Only the diode D_1 starts conducting and thus current flows from diode D_1 and it appear across the load R_L. As a result, the positive half cycle of the input is appeared at the load R_L.

During the negative half cycle of the input, diode D_2 is connected with lower end (B) and diode D_1 is connected with upper end (A). The diode D_1 is in reverse bias and does not conduct, only diode D_2 starts conducting and thus current flows from diode D_2 and appears across the load R_L.

By comparing the current through the load R_L in the positive and negative half cycles, it can be concluded that the

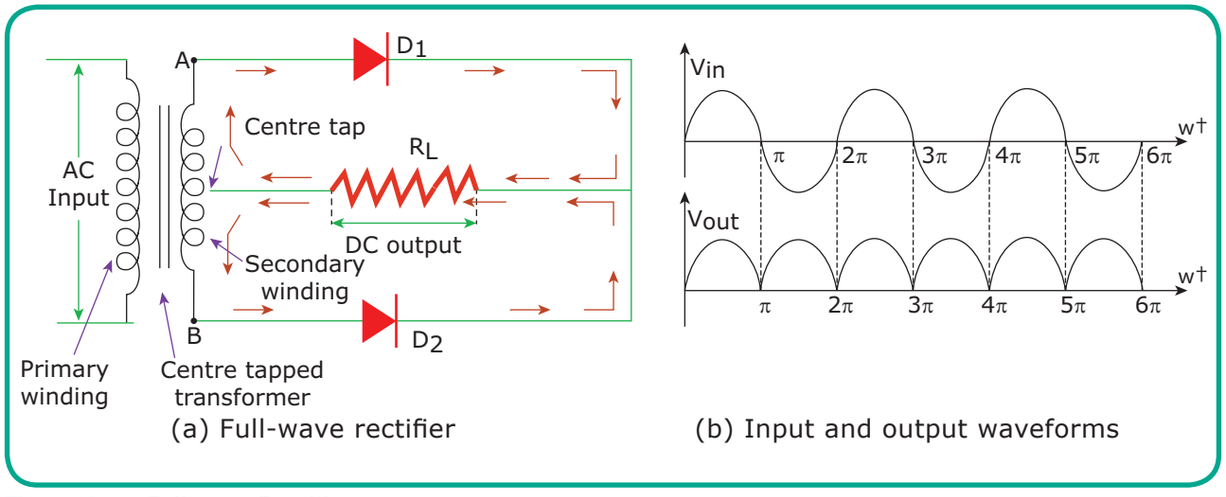


Figure 4.11 Full-wave Rectifier

ELEC_EQUIP_Chapter04.indd 89 11/10/2020 11:38:45 AM